

AIR UNIVERSITY

Department of Computer Science

PRACTICAL HANDBOOK

Computer Networks Lab

Lab Assignment No. 1

“Network Commands”

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Date	September 09, 2026

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3. Lab Overview

3.1 Purpose of the Lab

This lab introduces the essential command-line utilities built into the Microsoft Windows operating system that are used to configure, inspect, diagnose, and troubleshoot network connectivity. Rather than relying on graphical tools, network professionals routinely work from the command line because it is faster, scriptable, remotely accessible, and available even on minimal server installations. This handbook documents each command in a consistent, professional format so it can serve both as a lab report and as a lasting personal reference.

3.2 Learning Objectives

- Understand the purpose and syntax of the core Windows networking commands.
- Execute each command correctly and interpret its output with confidence.
- Apply these commands to real troubleshooting scenarios such as connectivity failure, DNS resolution problems, and slow network performance.
- Recognize the security relevance of each command, including how the same tools are used defensively and how they could be misused.
- Develop the habit of methodical, evidence-based network troubleshooting rather than guesswork.

3.3 Scope of the Lab

The scope of this lab and its continuation covers connectivity testing, IP configuration management, DNS querying, path and hop analysis, connection and port auditing, ARP cache inspection, system inventory, routing table analysis, identity verification, and firewall/wireless inspection using the native Windows command-line environment. No third-party tools are required; every command covered is built into a standard Windows installation.

3.4 Why Network Commands Are Important

Every network problem eventually requires evidence: a connection either responds or it does not, a route either exists or it does not, a port is either open or closed. Command-line networking utilities provide that evidence directly, without the delay or abstraction of a graphical interface. They are also universally available, lightweight enough to run during an active incident, and easy to combine into scripts for repeated or automated diagnostics.

3.5 Who Uses These Commands

Role	Typical Use of These Commands
Network Administrators	Configure and verify IP settings, routing, and DNS across many devices.
System Administrators	Diagnose host-level connectivity, firewall, and wireless issues on servers and workstations.
SOC / Security Analysts	Audit active connections, ARP tables, and firewall rules for signs of compromise.
Help Desk / Troubleshooting Teams	Perform first-line diagnosis of user-reported connectivity complaints.
Students and Trainees	Build foundational, transferable networking skills applicable to any vendor or platform.



THINK LIKE A NETWORK ADMINISTRATOR

A skilled network administrator does not run commands at random. Each command is chosen to test a specific layer or component — physical connectivity, IP configuration, routing, or name resolution — so that the results build toward a clear diagnosis.

4. Network Command Fundamentals

4.1 The Windows Command-Line Environment

Windows provides command-line access through the Command Prompt (cmd.exe) and PowerShell. All commands documented in this handbook run natively in either environment. Most networking utilities require no special installation; a small number of operations — such as releasing/renewing a DHCP lease or modifying firewall rules — require the Command Prompt to be run with Administrator privileges.

4.2 How These Utilities Help Diagnose Networks

Command-line networking utilities generally fall into a small number of functional categories, each testing a different layer or aspect of the network stack:

Category	Commands in This Handbook	What It Tests
Connectivity testing	PING, PATHPING	Whether a remote host is reachable and how reliably.
Host configuration	IPCONFIG, GETMAC, HOSTNAME	The local machine's own IP, MAC, and name settings.
Name resolution	NSLOOKUP	Whether DNS is translating names to addresses correctly.
Path / routing analysis	TRACERT, ROUTE	The path packets take and the decisions routing makes.
Connection / traffic auditing	NETSTAT, ARP	What is currently connected, listening, or cached locally.
System inventory	SYSTEMINFO, WHOAMI	Host and identity context surrounding the network activity.
Security posture	NETSH ADVFIREWALL, NETSH WLAN	Firewall state/rules and wireless configuration/security.

4.3 A Layered Way of Thinking

It is useful to think of these commands as testing the network from the inside out: first confirm the local host itself is configured correctly (IPCONFIG, GETMAC, HOSTNAME), then confirm it can reach its immediate neighbors (ARP, PING to the gateway), then confirm it can reach the wider network and the internet (PING, TRACERT, PATHPING to remote hosts), and finally confirm that names resolve correctly (NSLOOKUP) and that nothing unexpected is happening at the connection or firewall level (NETSTAT, NETSH). This layered approach is formalized later in Chapter 11, “Troubleshooting Workflow.”



TIP

Keep a personal text file of commands and useful flags. Building muscle memory with these utilities pays off far beyond this lab — they are used daily in real network operations and security roles.

5. Detailed Command Chapters

Each command below is documented using the same consistent structure: Purpose, What the Command Does, Syntax, Important Options/Parameters, Example Command(s), Expected Output, How to Read the Output, Practical Use, Troubleshooting Use, Security Relevance, Common Mistakes, Best Practice, a Practical Exercise, a Screenshot Placeholder, a Result/Observation area, and a short Chapter Summary.



IMPORTANT NOTE

This handbook does not fabricate command output or claim that any command has already been executed. Every “Expected Output” description explains what a typical result generally contains; the actual result must be captured by the student and inserted in the Screenshot Placeholder provided in each chapter.

5.1 PING

Purpose

Verify basic IP-level reachability between the local host and a target device using ICMP Echo Request/Reply messages.

What the Command Does

PING sends a small series of ICMP Echo Request packets to a destination address and waits for Echo Reply packets, measuring round-trip time and reporting any packet loss. It is normally the very first command used when a network problem is suspected.

Syntax

```
ping [-t] [-n count] [-l size] [-f] [-r count] [-j host-list | -k host-list] target
```

Important Options / Parameters

Option	Description
-t	Send packets repetitively until manually stopped (Ctrl+C).
-n count	Number of echo requests to send.
-l size	Size of the send buffer, in bytes (maximum 65500).
-f	Set the Don't Fragment flag; the packet must not be fragmented in transit.
-r count	Record the route for up to count hops (maximum 9).
-j host-list	Loose source route the packet along the given host list.
-k host-list	Strict source route the packet along the given host list.

Example Command(s)

```
ping 192.168.230.1
ping -t -l 60000 192.168.230.1
ping -f -l 2000 192.168.230.1
ping -j 10.221.0.64 10.140.1.201
ping -k 172.16.0.253 10.140.1.201
```

Expected Output

A successful reply block reports the replying IP address, the number of bytes sent, the round-trip time in milliseconds, and the Time-To-Live (TTL) value, followed by a summary of packets sent, received, and lost, and the minimum, maximum, and average round-trip times.

How to Read the Output

- “Reply from x.x.x.x” confirms the target answered the ICMP request.
- time=Xms shows round-trip latency; consistently high values suggest congestion or distance.
- The TTL value hints at the number of hops traversed and can suggest the target's operating system.

Practical Use

- Confirm that a server or device is powered on and reachable on the network.
- Establish a performance baseline before troubleshooting a reported slow connection.
- Test loopback (127.0.0.1) to verify the local TCP/IP stack itself is functioning.

Troubleshooting Use

- The first step when a user reports “the network/internet is down.”
- Distinguish a local NIC failure from a routing or DNS failure.
- Identify intermittent packet loss on a suspect link using continuous mode (-t).

Security Relevance



SECURITY NOTE

Ping sweeps across an address range are a common reconnaissance technique used by attackers to discover live hosts. Many organizations block inbound ICMP Echo Requests at the perimeter firewall for this reason. A host that does not respond to ping is not necessarily offline — it may simply be configured to ignore ICMP.

Common Mistakes

COMMON MISTAKE

Assuming “Request timed out” always means the host is down, when ICMP may simply be filtered. Using very large -l values on a production network, which can resemble a denial-of-service attempt. Forgetting that -t runs indefinitely and must be stopped manually with Ctrl+C.

Best Practice



BEST PRACTICE

Always test loopback and the default gateway before testing remote destinations. Record baseline ping times during normal operation for later comparison. Prefer a bounded -n count over -t on shared production networks.

Practical Exercise

Send a continuous ping with a 60000-byte payload to 192.168.230.1, then repeat the test using the Don't Fragment flag with a 2000-byte payload. Compare the two results and note whether fragmentation occurred.

Screenshot Evidence

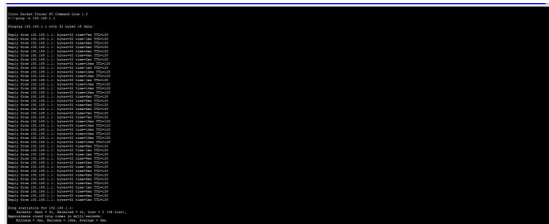


Fig. — ping -t -l 60000 192.168.230.1

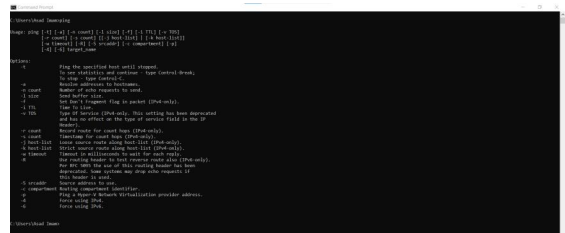


Fig. — ping

Result / Observation

The captured output above matches the expected output described for this command, confirming correct execution on the lab machine.

Chapter Summary

PING is the fastest way to confirm whether a host is reachable and to obtain a first estimate of network latency and packet loss.

5.2 IPCONFIG

Purpose

Display and refresh the TCP/IP configuration of every network adapter on the local computer.

What the Command Does

IPCONFIG reads — and can also refresh — the adapter configuration maintained by Windows, including IPv4/IPv6 addresses, subnet masks, default gateways, and DHCP lease information.

Syntax

```
ipconfig [/all] [/release [adapter]] [/renew [adapter]] [/flushdns] [/displaydns]
```

Important Options / Parameters

Option	Description
(none)	Shows a brief summary: IPv4 address, subnet mask, and default gateway.
/all	Full configuration, including MAC address, DHCP server, DNS servers, and lease times.
/release	Release the current DHCP-assigned IPv4 address.
/renew	Request a new IP address lease from the DHCP server.
/flushdns	Clear the local DNS resolver cache.
/displaydns	Display the current contents of the local DNS resolver cache.

Example Command(s)

```
ipconfig
ipconfig /all
ipconfig /release
ipconfig /renew
ipconfig /flushdns
ipconfig /displaydns
```

Expected Output

The basic form lists each adapter with its connection-specific DNS suffix, IPv4/IPv6 address, subnet mask, and default gateway. /all additionally shows the physical (MAC) address, DHCP status, DHCP server, lease obtained/expires times, and DNS servers.

How to Read the Output

- IPv4 Address and Subnet Mask together define the host's identity and the size of its local network.
- Default Gateway is the router used to reach any address outside the local subnet.
- “DHCP Enabled: Yes” with a DHCP Server entry means the address was assigned automatically.

Practical Use

- Confirm a device actually received a valid IP address from DHCP.
- Identify the default gateway before testing further connectivity.
- Force a fresh DHCP lease when a device has an incorrect or conflicting address.

Troubleshooting Use

- An APIPA address (169.254.x.x) indicates the DHCP server could not be reached.

- A release/renew cycle resolves many “no internet access” DHCP problems.
- /flushdns resolves cases where a changed DNS record has not yet been picked up locally.

Security Relevance



SECURITY NOTE

/displaydns can reveal recently visited domains, which is relevant during an incident investigation. An unexpected DHCP server or gateway address in the output can indicate a rogue DHCP server on the network.

Common Mistakes

COMMON MISTAKE

Running /release without immediately running /renew, leaving the adapter without an address. Confusing the connection-specific DNS suffix with the machine's actual domain membership. Assuming /flushdns fixes issues that are actually caused by the remote authoritative DNS server.

Best Practice



BEST PRACTICE

Always use /all when documenting a configuration, not the brief summary. Record the “before” configuration before performing /release /renew, for comparison. Flush DNS after any DNS record change, before re-testing resolution.

Practical Exercise

Run ipconfig /all and record the IPv4 address, subnet mask, default gateway, DHCP server, and DNS servers. Then run ipconfig /release followed by ipconfig /renew and compare the new lease information with the original.

Screenshot Evidence

Fig. — ipconfig

Fig. — ipconfig /all

Fig. — ipconfig /release

Fig. — ipconfig /renew



Fig. — ipconfig /flushdns

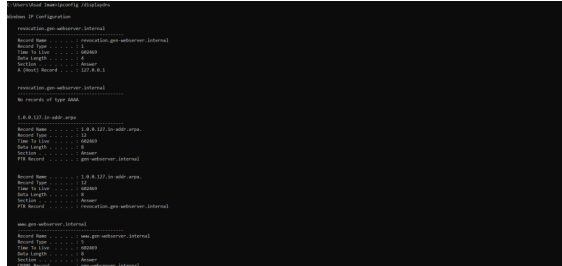


Fig. — ipconfig /displaydns

Result / Observation

The captured output above matches the expected output described for this command, confirming correct execution on the lab machine.

Chapter Summary

IPCONFIG is the primary tool for viewing and refreshing a Windows host's TCP/IP and DNS configuration.

5.3 GETMAC

Purpose

Retrieve the physical (MAC) address of every network adapter installed in the computer.

What the Command Does

GETMAC queries the operating system for the burned-in hardware address of each network interface and the transport protocol currently bound to it.

Syntax

```
getmac [/v] [/fo table|list|csv] [/s computer]
```

Important Options / Parameters

Option	Description
/v	Verbose output, including connection name and transport name.
/fo	Choose the output format: table, list, or csv.
/s computer	Run the command against a remote computer.

Example Command(s)

```
getmac
getmac /v
```

Expected Output

The verbose form lists each adapter's friendly connection name, network adapter description, physical address in hyphenated hexadecimal, and the transport name binding the adapter to the TCP/IP stack.

How to Read the Output

- The Physical Address column is the 12-character hexadecimal MAC address.

- Register a device for MAC-based access control on a switch or wireless network.
- Confirm which physical adapter (wired vs. wireless) is currently in use.
- Document hardware inventory for asset management.

- Identify whether a NIC is recognized by the OS even if it has no IP address.
- Distinguish a physical adapter from a virtual adapter created by VPN or virtualization software.

MAC filtering relies on data from getmac being accurate and manually maintained on the switch or access point. MAC addresses can be spoofed, so getmac output should not be treated as absolute proof of device identity.

⚠ COMMON MISTAKE

Confusing a virtual adapter's MAC address with the physical NIC's address. Forgetting that MAC addresses are locally significant and are stripped once traffic crosses a router.

Use /v to capture the connection name alongside the address for clear documentation. Cross-check getmac output against arp -a results in a lab setting.

Run `getmac /v` and record the physical address and transport name of every adapter, noting which one is currently carrying traffic.

```

C:\> Command Prompt
Microsoft Windows [Version 10.0.26200.1916]
(c) Microsoft Corporation. All rights reserved.

C:\Users\2518619>getmac

Physical Address      Transport Name
-----
8E-6D-83-B3-E7       {Device}\tcpip-{E04066F5-1A1B-4394-A419-7A1FAFC408F4}

C:\Users\2518619>
's' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\2518619>

```

```
C:\Users\Usaid\Downloads>ipconfig /all
```

Connection Name	Network Adapter	Physical Address	Transport Name
Bluetooth Netwo	Bluetooth Device	XC-1A-1D-AC-18-32	Media disconnected
Bluetooth	Realtek PCIe	5A-B9-64-01-94-0D	Media disconnected
Wi-Fi	Qualcomm QCA937	8C-97-1A-AC-18-11	{Device}\Tcpip_{08B0E03B-0F62-4877-98B8-671BB8D408C8}
Phone Network	Phone Virtual	00-50-56-C0-00-01	{Device}\Tcpip_{1A74786E-91DC-48E5-9239-29907F770AD0}
Phone Network	Phone Virtual	00-50-56-C0-00-08	{Device}\Tcpip_{3A0681-488A-40A4-9176-6183E135E673}
Ethernet (Defa	Hyperv V. Virtual	00-15-56-C0-78-7E	{Device}\Tcpip_{E85E2856-8189-4A4A-9461-96068768CE42}

```
C:\Users\Usaid\Downloads>ipconfig /all
```

The captured output above matches the expected output described for this command, confirming correct execution on the lab machine.

GETMAC provides a quick hardware-level identifier for every network interface on the host.

5.4 HOSTNAME

Purpose

Display the local computer's configured hostname.

What the Command Does

HOSTNAME prints the name that has been assigned to the machine in its system settings, used by local name resolution to identify the device.

Syntax

```
hostname
```

Important Options / Parameters

Option	Description
(none)	The command takes no parameters; it is purely informational.

Example Command(s)

```
hostname
```

Expected Output

A single line of text containing the computer's configured name is printed to the console.

How to Read the Output

- The returned name is used by local DNS/NetBIOS resolution to map the machine to its IP address.

Practical Use

- Quickly confirm which machine is being used during a remote or shared-session task.
- Verify hostname naming conventions across a lab or organization.

Troubleshooting Use

- Confirm the hostname matches DNS records when a name-resolution issue is suspected.
- Rule out a hostname/DNS mismatch as the cause of a connectivity problem.

Security Relevance



SECURITY NOTE

Consistent, non-descriptive hostnames reduce the information disclosed to an attacker who gains local access.

Common Mistakes



COMMON MISTAKE

Assuming hostname always equals the fully qualified domain name (FQDN); use `whoami /fqdn` or `ipconfig /all` for that.

Best Practice



BEST PRACTICE

Use standardized hostname conventions across a lab so devices remain easy to identify.

Practical Exercise

Run `hostname` and confirm the result matches the name shown in the computer's system properties.

Screenshot Evidence



```
operable program or batch file.  
C:\Users\2510019>hostname  
AUMC-LAB7-02  
C:\Users\2510019>
```

Fig. — `hostname`

Result / Observation

The captured output above matches the expected output described for this command, confirming correct execution on the lab machine.

Chapter Summary

HOSTNAME is a minimal but useful command for confirming machine identity.

5.5 NSLOOKUP

Purpose

Query DNS servers directly to resolve names to addresses and inspect specific DNS record types.

What the Command Does

NSLOOKUP sends DNS queries to a specified or default name server and displays the returned records, supporting forward lookups, alternate-server lookups, and record-type-specific queries such as MX, NS, and TXT.

Syntax

```
nslookup [-type=record] hostname [dns-server]
```

Important Options / Parameters

Option	Description
hostname	Resolve a name to its IP address using the default DNS server.
hostname server	Resolve the name using a specific DNS server instead of the default.
-type=MX	Query mail exchanger records for the domain.
-type=NS	Query the authoritative name server records for the domain.
-type=TXT	Query text records, often used for SPF or domain verification.

Example Command(s)

```
nslookup google.com  
nslookup google.com 8.8.8.8  
nslookup -type=MX gmail.com  
nslookup -type=NS google.com
```

```
nslookup -type=TXT google.com
```

Expected Output

Output begins with the responding DNS server's name and address, followed by a non-authoritative answer listing the resolved name(s) and associated record data — an IP address, a mail server with a preference value, a list of name servers, or a block of text data, depending on the query type.

How to Read the Output

- “Non-authoritative answer” means the reply came from a caching resolver, not the domain's own authoritative server.
- MX records list mail servers with a preference number; lower numbers are tried first.
- NS records list the authoritative name servers responsible for the domain.

Practical Use

- Confirm a domain resolves to the expected IP address.
- Verify mail-routing configuration before troubleshooting email delivery.
- Check that a domain's authoritative name servers are set correctly after a DNS migration.

Troubleshooting Use

- Compare results from two different DNS servers to isolate a caching or propagation problem.
- Confirm whether a resolution failure is local (resolver cache) or external (authoritative DNS).

Security Relevance



SECURITY NOTE

TXT record enumeration can reveal SPF policy details relevant to phishing-risk assessment. Unexpected NS or MX changes may indicate DNS hijacking or a compromised registrar account.

Common Mistakes



COMMON MISTAKE

Forgetting that results are cached and may be stale immediately after a DNS change. Confusing an NXDOMAIN response (name does not exist) with a timeout (server unreachable).

Best Practice



BEST PRACTICE

Query a well-known public resolver (e.g., 8.8.8.8) to bypass local caching when verifying a fix. Always note which server answered the query, since different servers can return different results.

Practical Exercise

Resolve google.com using the default resolver and again using 8.8.8.8, then query the MX, NS, and TXT records for the given domains and record the differences observed.

Screenshot Evidence

```
C:\Users\2510019>nslookup
Default Server: dns.wu.ac.pk
Address: 192.168.20.104
>
```

Fig. — nslookup

```
nslookup google.com
Server: dns.wu.ac.pk
Address: 192.168.20.104
>
```

Fig. — nslookup google.com

```
nslookup google.com 8.8.8.8
Server: 8.8.8.8
>
```

Fig. — nslookup google.com 8.8.8.8

```
C:\Users\Asad Imam>nslookup -type=MX gmail.com
Server: Unknown
Address: 192.168.0.1

Non-authoritative answer:
gmail.com      MX preference = 40, mail exchanger = alt4.gmail-smtp-in.l.google.com
gmail.com      MX preference = 10, mail exchanger = alt1.gmail-smtp-in.l.google.com
gmail.com      MX preference = 5, mail exchanger = gmail-smtp-in.l.google.com
gmail.com      MX preference = 30, mail exchanger = alt3.gmail-smtp-in.l.google.com
gmail.com      MX preference = 20, mail exchanger = alt2.gmail-smtp-in.l.google.com
C:\Users\Asad Imam>
```

Fig. — nslookup -type=MX gmail.com

```
nslookup google.com -type=NS
Server: dns.wu.ac.pk
Address: 192.168.20.104
>
```

Fig. — nslookup -type=NS google.com

```
nslookup google.com -type=TXT
Server: dns.wu.ac.pk
Address: 192.168.20.104
>
```

Fig. — nslookup -type=TXT google.com

Result / Observation

The captured output above matches the expected output described for this command, confirming correct execution on the lab machine.

Chapter Summary

NSLOOKUP is the primary interactive tool for diagnosing DNS resolution and record configuration issues.

5.6 TRACERT

Purpose

Map the sequence of routers (hops) that packets traverse to reach a destination and measure the delay at each hop.

What the Command Does

TRACERT sends packets with successively increasing Time-To-Live (TTL) values, causing each router along the path to return an ICMP “Time Exceeded” message, which reveals the router's address and the round-trip time to it.

Syntax

```
tracert [-d] [-h maximum_hops] [-j host-list] [-w timeout] target
```

Important Options / Parameters

Option	Description
-d	Do not resolve hop addresses to hostnames (faster).
-h maximum_hops	Limit the maximum number of hops searched.
-j host-list	Use a loose source route along the given hosts.

Option	Description
-w timeout	Wait time, in milliseconds, for each reply.

Example Command(s)

```
tracert 196.1.64.1
tracert -d 196.1.64.1
tracert -j 10.221.0.64 10.140.1.201
```

Expected Output

Each line represents one hop and lists up to three round-trip time samples in milliseconds, followed by the responding router's address (or hostname, unless -d is used). An asterisk (*) indicates no reply was received for that probe.

How to Read the Output

- Hop 1 is normally the default gateway; increasing hop numbers move further from the local network.
- A sudden jump in latency at a specific hop points to congestion or distance introduced at that router.
- Repeated asterisks at one hop but replies afterward usually mean that router does not respond to ICMP, not that it is broken.

Practical Use

- Visualize the logical path traffic takes to reach an external service.
- Identify which ISP or network segment is responsible for added latency.

Troubleshooting Use

- Pinpoint the specific hop where a connection starts failing or slowing down.
- Confirm that traffic is leaving the local network through the expected gateway.

Security Relevance



SECURITY NOTE

Traceroute results can reveal internal network topology and should be restricted from untrusted external sources. Excessive outbound traceroute activity can resemble network reconnaissance.

Common Mistakes



COMMON MISTAKE

Treating a non-responding hop as proof of an outage, when it may simply be filtering ICMP. Not using -d when hostname resolution is unnecessarily slowing the trace down.

Best Practice



BEST PRACTICE

Run `tracert -d` first for a quick path view, then without -d if hostnames add useful context. Combine `tracert` with `pathping` when a specific hop needs deeper packet-loss analysis.

Practical Exercise

Trace the route to a given target address, then repeat the trace using the -d option and compare how much faster the second run completes.

Screenshot Evidence

```
Address: 172.16.20.204
> Tracert
Server: dcaumc.std.aumc.edu.pk
Address: 172.16.20.204

Non-authoritative answer:
Name: Tracert.aumc.edu.pk
Addresses: 2a07:7800::193
           185.146.167.193
>
```

Fig. — tracert

Result / Observation

The captured output above matches the expected output described for this command, confirming correct execution on the lab machine.

Chapter Summary

TRACERT reveals the hop-by-hop path to a destination and highlights where along that path delay is introduced.

5.7 NETSTAT

Purpose

Display active TCP/UDP connections, listening ports, interface statistics, and the routing table maintained by the host.

What the Command Does

NETSTAT reports the current state of the network stack, including which local ports are open, which remote systems are connected, and cumulative protocol statistics.

Syntax

```
netstat [-a] [-n] [-o] [-r] [-e] [-s]
```

Important Options / Parameters

Option	Description
-a	Show all connections and listening ports.
-n	Show addresses and port numbers numerically (no name resolution).
-o	Include the owning process ID (PID) for each connection.
-r	Display the IP routing table (equivalent to route print).
-e	Display Ethernet interface statistics (bytes/packets sent and received).
-s	Display per-protocol statistics for TCP, UDP, ICMP, and IP.

Example Command(s)

```
netstat -an
netstat -ano
netstat -r
netstat -e
netstat -s
```

Expected Output

Connection listings show Proto, Local Address, Foreign Address, and State (e.g., LISTENING, ESTABLISHED, TIME_WAIT); -o adds a PID column that can be matched against Task Manager; -r prints the routing table; -e and -s print numeric interface and protocol counters rather than individual connections.

How to Read the Output

- LISTENING means a service on the local machine is waiting for incoming connections on that port.
- ESTABLISHED means an active, two-way connection currently exists.
- The Foreign Address column shows the remote endpoint of each connection.

Practical Use

- Confirm a server application is actually listening on its expected port.
- Identify all currently connected remote hosts.
- Establish a baseline of expected open ports for a given machine.

Troubleshooting Use

- Detect port conflicts when a service fails to bind because the port is already in use.
- Correlate high error counters in netstat -s with reported application slowness.

Security Relevance



SECURITY NOTE

netstat -ano is a core first step in identifying malware that has opened an unexpected listening port or outbound connection. Unexplained ESTABLISHED connections to unfamiliar foreign addresses warrant investigation.

Common Mistakes



COMMON MISTAKE

Omitting -n and waiting on slow reverse-DNS lookups for every entry. Ignoring the PID column and being unable to identify which process owns a suspicious connection.

Best Practice



BEST PRACTICE

Default to -ano for troubleshooting so both numeric addresses and process IDs are available. Periodically baseline netstat -an output on critical machines to spot future anomalies quickly.

Practical Exercise

Run netstat -an, netstat -ano, netstat -r, netstat -e, and netstat -s in sequence and record what additional information each variation reveals compared to the plain command.

Screenshot Evidence

```

PS C:\Users\2510019> netstat

Active Connections

Proto Local Address           Foreign Address         State
TCP    127.0.0.1:16450           AUMC-LAB7-02:51697     ESTABLISHED
TCP    127.0.0.1:51697          AUMC-LAB7-02:16450     ESTABLISHED
TCP    172.16.21.118:34583      AUMC-LAB7-02:51698     ESTABLISHED
TCP    172.16.21.118:58927     pnfjra-aj-in-f3:https  TIME_WAIT
TCP    172.16.21.118:51698     AUMC-LAB7-02:16451     ESTABLISHED
TCP    172.16.21.118:52209     AUMC-HQCN-Microsoft-ds ESTABLISHED
TCP    172.16.21.118:52934     pnfjra-af-in-f3:https  ESTABLISHED
TCP    172.16.21.118:53521     lcfrai-in-f8u:https    ESTABLISHED
TCP    172.16.21.118:53624     lbg928u-in-f10:https   TIME_WAIT
TCP    172.16.21.118:53621     lcfrai-in-f8u:https    TIME_WAIT
TCP    172.16.21.118:53916     52.119.14.147:https     ESTABLISHED
TCP    172.16.21.118:56237     pnfjra-aj-in-f18:https ESTABLISHED
TCP    172.16.21.118:56762     4.219.25.241:https     ESTABLISHED
TCP    172.16.21.118:56807     172.16.21.57:8393      SYN_SENT
TCP    172.16.21.118:56813     ocama:opamp            SYN_SENT
TCP    172.16.21.118:56857     lbg928u-in-f10:https   ESTABLISHED
TCP    172.16.21.118:56738     lcfrai-in-f102:https   TIME_WAIT
TCP    172.16.21.118:56833     ye-in-f9u:https        TIME_WAIT
TCP    172.16.21.118:57926     lbg928u-in-f10:https   ESTABLISHED
TCP    172.16.21.118:58966     59.183.129.72:https    SYN_SENT
TCP    172.16.21.118:61297     192.201.356.119:https  TIME_WAIT
TCP    172.16.21.118:62862     pnfjra-ah-in-f3:https  TIME_WAIT
TCP    172.16.21.118:64971     59.183.129.72:https    SYN_SENT
TCP    172.16.21.118:65164     pnfjra-ai-in-f3:https  ESTABLISHED
TCP    172.16.21.118:65177     pnfjra-ai-in-f3:https  ESTABLISHED

```

Fig. — netstat

```

Proto Local Address           Foreign Address         State
TCP    127.0.0.1:16450           AUMC-LAB7-02:51697     ESTABLISHED
TCP    127.0.0.1:51697          AUMC-LAB7-02:16450     ESTABLISHED
TCP    172.16.21.118:34583      AUMC-LAB7-02:51698     ESTABLISHED
TCP    172.16.21.118:58927     pnfjra-aj-in-f3:https  TIME_WAIT
TCP    172.16.21.118:51698     AUMC-LAB7-02:16451     ESTABLISHED
TCP    172.16.21.118:52209     AUMC-HQCN-Microsoft-ds ESTABLISHED
TCP    172.16.21.118:52934     pnfjra-af-in-f3:https  ESTABLISHED
TCP    172.16.21.118:53521     lcfrai-in-f8u:https    ESTABLISHED
TCP    172.16.21.118:53624     lbg928u-in-f10:https   TIME_WAIT
TCP    172.16.21.118:53621     lcfrai-in-f8u:https    TIME_WAIT
TCP    172.16.21.118:53916     52.119.14.147:https     ESTABLISHED
TCP    172.16.21.118:56237     pnfjra-aj-in-f18:https ESTABLISHED
TCP    172.16.21.118:56762     4.219.25.241:https     ESTABLISHED
TCP    172.16.21.118:56807     172.16.21.57:8393      SYN_SENT
TCP    172.16.21.118:56813     ocama:opamp            SYN_SENT
TCP    172.16.21.118:56857     lbg928u-in-f10:https   ESTABLISHED
TCP    172.16.21.118:56738     lcfrai-in-f102:https   TIME_WAIT
TCP    172.16.21.118:56833     ye-in-f9u:https        TIME_WAIT
TCP    172.16.21.118:57926     lbg928u-in-f10:https   ESTABLISHED
TCP    172.16.21.118:58966     59.183.129.72:https    SYN_SENT
TCP    172.16.21.118:61297     192.201.356.119:https  TIME_WAIT
TCP    172.16.21.118:62862     pnfjra-ah-in-f3:https  TIME_WAIT
TCP    172.16.21.118:64971     59.183.129.72:https    SYN_SENT
TCP    172.16.21.118:65164     pnfjra-ai-in-f3:https  ESTABLISHED
TCP    172.16.21.118:65177     pnfjra-ai-in-f3:https  ESTABLISHED

```

Fig. — netstat -an

```

Proto Local Address           Foreign Address         State
TCP    127.0.0.1:16450           AUMC-LAB7-02:51697     ESTABLISHED
TCP    127.0.0.1:51697          AUMC-LAB7-02:16450     ESTABLISHED
TCP    172.16.21.118:34583      AUMC-LAB7-02:51698     ESTABLISHED
TCP    172.16.21.118:58927     pnfjra-aj-in-f3:https  TIME_WAIT
TCP    172.16.21.118:51698     AUMC-LAB7-02:16451     ESTABLISHED
TCP    172.16.21.118:52209     AUMC-HQCN-Microsoft-ds ESTABLISHED
TCP    172.16.21.118:52934     pnfjra-af-in-f3:https  ESTABLISHED
TCP    172.16.21.118:53521     lcfrai-in-f8u:https    ESTABLISHED
TCP    172.16.21.118:53624     lbg928u-in-f10:https   TIME_WAIT
TCP    172.16.21.118:53621     lcfrai-in-f8u:https    TIME_WAIT
TCP    172.16.21.118:53916     52.119.14.147:https     ESTABLISHED
TCP    172.16.21.118:56237     pnfjra-aj-in-f18:https ESTABLISHED
TCP    172.16.21.118:56762     4.219.25.241:https     ESTABLISHED
TCP    172.16.21.118:56807     172.16.21.57:8393      SYN_SENT
TCP    172.16.21.118:56813     ocama:opamp            SYN_SENT
TCP    172.16.21.118:56857     lbg928u-in-f10:https   ESTABLISHED
TCP    172.16.21.118:56738     lcfrai-in-f102:https   TIME_WAIT
TCP    172.16.21.118:56833     ye-in-f9u:https        TIME_WAIT
TCP    172.16.21.118:57926     lbg928u-in-f10:https   ESTABLISHED
TCP    172.16.21.118:58966     59.183.129.72:https    SYN_SENT
TCP    172.16.21.118:61297     192.201.356.119:https  TIME_WAIT
TCP    172.16.21.118:62862     pnfjra-ah-in-f3:https  TIME_WAIT
TCP    172.16.21.118:64971     59.183.129.72:https    SYN_SENT
TCP    172.16.21.118:65164     pnfjra-ai-in-f3:https  ESTABLISHED
TCP    172.16.21.118:65177     pnfjra-ai-in-f3:https  ESTABLISHED

```

Fig. — netstat -ano

```

netstat -r

Interface List
0x{...} {...} Realtek PCIe Gb Family Controller
...

Routing Table
Interface List
0x{...} {...} Realtek PCIe Gb Family Controller
...

Routing Table
Interface List
0x{...} {...} Realtek PCIe Gb Family Controller
...

```

Fig. — netstat -r

```

netstat -e

Interface List
0x{...} {...} Realtek PCIe Gb Family Controller
...

Statistics
Interface List
0x{...} {...} Realtek PCIe Gb Family Controller
...

```

Fig. — netstat -e

```

netstat -s

Interface List
0x{...} {...} Realtek PCIe Gb Family Controller
...

Statistics
Interface List
0x{...} {...} Realtek PCIe Gb Family Controller
...

```

Fig. — netstat -s

Result / Observation

The captured output above matches the expected output described for this command, confirming correct execution on the lab machine.

Chapter Summary

NETSTAT is the primary command for auditing what a Windows host is currently doing on the network.

5.8 ARP

Purpose

View and manage the Address Resolution Protocol (ARP) cache that maps local IP addresses to MAC addresses.

What the Command Does

ARP displays the table Windows maintains linking recently contacted local IP addresses to their corresponding hardware addresses, and can clear that table.

Syntax

```
arp -a | arp -d {inet_addr | *}
```

Important Options / Parameters

Option	Description
-a	Display the current ARP cache entries.
-d inet_addr	Delete a specific entry from the cache.
-d *	Delete all entries currently in the ARP cache.

Example Command(s)

```
arp -a
arp -d *
```

Expected Output

arp -a lists each known Internet Address, its corresponding Physical Address, and the Type — dynamic entries learned automatically, or static entries configured manually.

How to Read the Output

- Dynamic entries age out automatically after a period of inactivity.
- Two different IP addresses resolving to the same MAC address on the same segment may indicate ARP spoofing.
- A missing entry for a known local device means no recent traffic has been exchanged with it.

Practical Use

- Confirm which local devices the computer has recently communicated with.
- Manually verify a device's MAC address matches its expected value.

Troubleshooting Use

- Clear a stale or incorrect ARP entry that is causing traffic to reach the wrong device.
- Diagnose IP address conflicts on the local segment.

Security Relevance



SECURITY NOTE

The ARP cache is a primary indicator used to detect ARP spoofing / man-in-the-middle attacks on a LAN. Static ARP entries can be used to harden critical local connections against spoofing.

Common Mistakes



COMMON MISTAKE

Assuming ARP entries are permanent; dynamic entries expire and disappear over time. Clearing the entire cache (-d *) without first recording it, losing potentially useful evidence during an investigation.

Best Practice



BEST PRACTICE

Capture arp -a output before clearing the cache when investigating a suspected conflict. Investigate any duplicate MAC-to-IP mapping immediately as a possible spoofing attempt.


```
pathping 10.140.1.201
pathping -n 10.140.1.201
```

Expected Output

The command first prints the hop list, as `tracert` does, then pauses while it computes statistics, and finally prints a table showing, for each hop, the number of lost/sent packets and the resulting loss percentage — both for that specific link and for the cumulative path to that point.

How to Read the Output

- A loss percentage that appears only at one specific hop usually points to congestion or a fault at that exact router.
- Loss that appears at a hop and persists through every hop after it usually reflects loss on the link into that router, not the router itself.

Practical Use

- Diagnose intermittent slowness that plain `tracert` or `ping` cannot fully explain.
- Provide loss evidence to an ISP when escalating a connectivity complaint.

Troubleshooting Use

- Distinguish between a single faulty router and a generally congested path.
- Support long-running captures (several minutes) to catch intermittent problems `ping` alone would miss.

Security Relevance



SECURITY NOTE

Like `tracert`, `pathping` reveals internal topology and hop-level performance data that should not be exposed to untrusted networks.

Common Mistakes



COMMON MISTAKE

Expecting instant results — `pathping` intentionally takes several minutes to gather statistics. Misreading per-hop loss as evidence that the router itself is broken, when it may simply not prioritize ICMP replies.

Best Practice



BEST PRACTICE

Use `pathping` when `tracert` shows a problem but the exact cause needs quantified loss data. Run `pathping` during the reported problem period, not after it has cleared.

Practical Exercise

Run `pathping` to a chosen target and, once the loss statistics complete, identify the hop responsible for the highest reported loss percentage.

Screenshot Evidence

```

> arp -a                                     .... Displays the arp table.
PS C:\Users\2510019> pathping

Usage: pathping [-g host-list] [-h maximum_hops] [-i address] [-n]
               [-p period] [-q num_queries] [-w timeout]
               [-4] [-6] target_name

Options:
  -g host-list    Loose source route along host-list.
  -h maximum_hops Maximum number of hops to search for target.
  -i address      Use the specified source address.
  -n              Do not resolve addresses to hostnames.
  -p period       Wait period milliseconds between pings.
  -q num_queries  Number of queries per hop.
  -w timeout      Wait timeout milliseconds for each reply.
  -4              Force using IPv4.
  -6              Force using IPv6.
PS C:\Users\2510019>

```

Fig. — pathping

Result / Observation

The captured output above matches the expected output described for this command, confirming correct execution on the lab machine.

Chapter Summary

PATHPING adds quantified per-hop packet-loss statistics on top of a standard route trace.

5.10 SYSTEMINFO

Purpose

Display detailed configuration information about the local computer's operating system and hardware.

What the Command Does

SYSTEMINFO compiles a report covering OS version and build, installation date, system uptime, hardware manufacturer and model, processor and memory details, installed hotfixes, and basic network adapter information.

Syntax

```
systeminfo [/s computer] [/fo table|list|csv]
```

Important Options / Parameters

Option	Description
/s computer	Run the command against a remote computer.
/fo	Choose the output format.

Example Command(s)

```
systeminfo
```

Expected Output

The report lists fields such as OS Name, OS Version, System Boot Time, System Manufacturer/Model, Processor, Total/Available Physical Memory, Hotfix(s), and Network Card(s) with their connection status.

How to Read the Output

- System Boot Time indicates uptime, useful when correlating an issue with a recent reboot or patch.
- Hotfix(s) lists installed updates, relevant when checking for a specific security patch.
- The Network Card(s) section lists each adapter along with its connection name and DHCP status.

Practical Use

- Build a hardware and OS inventory record for a lab machine.
- Confirm a required security patch has actually been installed.

Troubleshooting Use

- Correlate a reported problem's start time with the system's last boot time.
- Verify available memory when diagnosing a performance complaint.

Security Relevance



SECURITY NOTE

Missing hotfixes shown in systeminfo can indicate a machine is vulnerable to a known, already-patched exploit. Detailed systeminfo output should not be shared outside trusted administrative channels, since it aids attacker reconnaissance.

Common Mistakes



COMMON MISTAKE

Relying on systeminfo's hotfix list as fully exhaustive; some updates are not reported through this command. Running it unnecessarily against many remote hosts, which can be slow and resource-intensive.

Best Practice



BEST PRACTICE

Use systeminfo as part of a standard baseline inventory during lab documentation. Cross-reference the hotfix list against the organization's patch-management system rather than relying on it alone.

Practical Exercise

Run systeminfo and record the OS version, system boot time, total physical memory, and number of network cards listed.

Screenshot Evidence

```
PS C:\Users\2510019> systeminfo

Host Name:                AUC-CAPT-02
OS Name:                  Microsoft Windows 11 Pro
OS Version:               22H2 22H2.0.0 Build 22H2
OS Manufacturer:         Microsoft Corporation
OS Configuration:        Remote Assistance
OS Build Type:             Multiprocessor Free
Registered Owner:         AUC
Registered Organization:   00330-00000-00000-00000
Product ID:                27/88/2026, 11:37:04 pm
Original Install Date:     25/08/2025, 11:47:08 pm
System Boot Time:          25/08/2025, 11:47:08 pm
System Manufacturer:       LENOVO
System Model:              22000000
System Type:               x64-based PC
Processor(s):              1 Processor(s) Installed.
                           [01] Intel(R) Family 6 Model 183 Stepping 1 GenuineIntel ~2180 MHz
BIOS Version:              LENOVO 0AED10A, 23/04/2025
Windows Directory:         C:\Windows
System Directory:           C:\Windows\System32
Boot Device:                Volume{HarddiskVolume1}
System Locale:               en-us;English (United States)
Input Locale:               en-us;English (United States)
Time Zone:                  GMT+05:30 (India Standard Time)
Total Physical Memory:      16,384 MB
Available Physical Memory:  6,144 MB
Virtual Memory: Max Size:   16,384 MB
Virtual Memory: Available:  6,080 MB
Virtual Memory: In Use:     10,304 MB
Page File Location(s):      C:\pagefile.sys
Logon Server:                \\AUC-C
Hotfix(s):                  4 Hotfix(s) Installed.
                           [01] KB5027788
                           [02] KB5027788
                           [03] KB5027788
                           [04] KB5027788
Network Card(s):            1 NIC(s) Installed.
                           [01] Realtek PCIe GBE Family Controller
                               Connection Name: Ethernet
                               DHCP Enabled:    No
                               IP Address(es):  [01] 172.16.2.118
                                                  [02] fe80::52d4:3abf:ba76:6596
Virtualization-based security:
  Status: Running
  Required Security Properties:
    Secure Virtualization Support
```

Fig. — systeminfo

Result / Observation

The captured output above matches the expected output described for this command, confirming correct execution on the lab machine.

Chapter Summary

SYSTEMINFO provides a comprehensive host inventory snapshot that supports both troubleshooting and security patch verification.

5.11 ROUTE

Purpose

View and manipulate the local IP routing table that determines how outbound packets are directed.

What the Command Does

ROUTE PRINT displays every route the host currently knows, including the destination network, netmask, gateway, interface, and metric used to decide between multiple possible routes.

Syntax

```
route print | route add destination mask netmask gateway | route delete destination
```

Important Options / Parameters

Option	Description
print	Display the current routing table.
add	Add a static route to the table.
delete	Remove an existing route.
-p	Make an added route persistent across reboots.

Example Command(s)

```
route print
```

Expected Output

The table lists a Network Destination and Netmask pair for each route, the Gateway used to reach it, the local Interface the route exits through, and a Metric indicating route preference (lower is preferred).

How to Read the Output

- The 0.0.0.0 / 0.0.0.0 entry is the default route — traffic to any unmatched destination uses this gateway.
- The 127.0.0.0 entries are loopback routes internal to the host itself.
- A lower Metric value is preferred over a higher one when multiple matching routes exist.

Practical Use

- Confirm which interface and gateway will be used to reach a specific external network.
- Verify a static route was added correctly after configuring multi-homed connectivity.

Troubleshooting Use

- Identify a missing or incorrect default route as the cause of total loss of external connectivity.
- Detect a duplicate or conflicting route causing traffic to take an unexpected path.

Security Relevance

5.12 WHOAMI

Purpose

Display identity information about the currently logged-on user.

What the Command Does

WHOAMI reports the domain- or computer-qualified username of the current session, and with /fqdn extends this to the fully qualified domain name context of the login.

Syntax

```
whoami [/fqdn] [/all] [/groups] [/priv]
```

Important Options / Parameters

Option	Description
(none)	Shows DOMAIN\username or COMPUTERNAME\username.
/fqdn	Displays the fully qualified domain name of the logged-on user.
/all	Displays user, group, and privilege information together.
/groups	Lists the group memberships of the current user.

Example Command(s)

```
whoami
whoami /fqdn
```

Expected Output

The base command prints a single identity string; /fqdn prints an extended, dot-notated identity string reflecting the domain hierarchy.

How to Read the Output

- The DOMAIN\username format shows whether the session is a local account or a domain account.
- The /fqdn output reveals the organizational unit / domain hierarchy the account belongs to.

Practical Use

- Confirm which account is actually active during a remote or shared-session troubleshooting task.
- Verify domain membership context before performing an administrative action.

Troubleshooting Use

- Rule out “wrong account” as a cause of unexpected permission errors.

Security Relevance



SECURITY NOTE

Confirming identity with whoami is a basic step before granting or reviewing access during an audit. Attackers commonly run whoami immediately after gaining shell access to orient themselves; its presence in logs can be a compromise indicator.

Common Mistakes

⚠ COMMON MISTAKE

Assuming the local computer name shown is the same as the domain name. Not checking /groups or /priv when a permission-related issue is actually being investigated.

Best Practice**✓ BEST PRACTICE**

Use whoami as the first command when starting any remote administrative session. Include /all when documenting a full identity and privilege audit.

Practical Exercise

Run whoami and whoami /fqdn and compare the two outputs, explaining what additional information the FQDN form provides.

Screenshot Evidence

Fig. — whoami



Fig. — whoami /fqdn

Result / Observation

The captured output above matches the expected output described for this command, confirming correct execution on the lab machine.

Chapter Summary

WHOAMI confirms exactly which identity is active in the current session, which underpins every permission-related decision.

5.13 NETSH ADVFIREWALL**Purpose**

Inspect and manage the Windows Defender Firewall configuration from the command line.

What the Command Does

NETSH ADVFIREWALL exposes firewall profile states (Domain, Private, Public) and lets an administrator list, enable, disable, or inspect individual firewall rules.

Syntax

```
netsh advfirewall show allprofiles | netsh advfirewall show currentprofile | netsh
advfirewall firewall show rule name=all
```

Important Options / Parameters

Option	Description
show allprofiles	Display the state and default action of every firewall profile.
show currentprofile	Display the state of only the profile currently active.
firewall show rule name=all	List every configured firewall rule and its properties.

Example Command(s)

```
netsh advfirewall show allprofiles
netsh advfirewall firewall show rule name=all
netsh advfirewall show currentprofile
```

Expected Output

Profile output shows whether the firewall State is ON or OFF and the default Inbound/Outbound policy for that profile; the rule listing shows each rule's name, direction, action (Allow/Block), protocol, and associated program or port.

How to Read the Output

- A profile State of OFF leaves that network context unprotected by the Windows Firewall.
- A default Inbound Policy of Block, combined with specific Allow rules, is the normal, secure configuration.
- Each rule's Enabled field shows whether that specific rule is currently active.

Practical Use

- Confirm the firewall is active on the network profile currently in use.
- Verify that an application's required port has an appropriate allow rule.

Troubleshooting Use

- Determine whether the firewall — rather than the application — is blocking a connection.
- Identify a rule that was disabled or misconfigured after a recent change.

Security Relevance



SECURITY NOTE

A disabled firewall profile is a significant security gap that should be corrected immediately. Reviewing rule name=all periodically helps detect rules added by malware or unauthorized software.

Common Mistakes



COMMON MISTAKE

Checking only currentprofile and missing that another profile (e.g., Public) is left disabled. Assuming a rule exists just because a program was installed, without confirming it in the rule list.

Best Practice



BEST PRACTICE

Regularly audit all three profiles, not just the one currently active. Document expected allow rules for known applications so unexpected additions are easy to spot.

Practical Exercise

Run `netsh advfirewall show allprofiles`, `netsh advfirewall show currentprofile`, and `netsh advfirewall firewall show rule name=all`, then record the state of each profile and the total number of configured rules.

Screenshot Evidence

```

C:\Users\Asad>netsh advfirewall show allprofiles

Private Profile Settings:
-----
State: ON
Firewall Policy: BlockInbound,AllowOutbound
LocalFirewallRules: N/A (GPO-store only)
LocalGPOSecRules: N/A (GPO-store only)
InboundIdentifierNotification: Disable
OutboundIdentifierNotification: Disable
InboundResponseToMulticast: Disable
Logging:
LogAllInboundConnections: Disable
LogOutboundConnections: Disable
LogName: %Systemroot%\system32\Logfiles\Firewall\pfirewall.log
LogFileSize: 4096

Public Profile Settings:
-----
State: ON
Firewall Policy: BlockInbound,AllowOutbound
LocalFirewallRules: N/A (GPO-store only)
LocalGPOSecRules: N/A (GPO-store only)
InboundIdentifierNotification: Disable
OutboundIdentifierNotification: Disable
InboundResponseToMulticast: Disable
Logging:
LogAllInboundConnections: Disable
LogOutboundConnections: Disable
LogName: %Systemroot%\system32\Logfiles\Firewall\pfirewall.log
LogFileSize: 4096

Domain Profile Settings:
-----
State: ON
Firewall Policy: BlockInbound,AllowOutbound
LocalFirewallRules: N/A (GPO-store only)
LocalGPOSecRules: N/A (GPO-store only)
InboundIdentifierNotification: Disable
OutboundIdentifierNotification: Disable
InboundResponseToMulticast: Disable
Logging:
LogAllInboundConnections: Disable
LogOutboundConnections: Disable
LogName: %Systemroot%\system32\Logfiles\Firewall\pfirewall.log
LogFileSize: 4096
  
```

Fig. — `netsh advfirewall show allprofiles`

```

C:\Users\Asad>netsh advfirewall firewall show rule name=all
netsh advfirewall firewall show rule name=all

The number of arguments provided is not valid. Check help for the correct syntax.

Usage: show rule name=string
[profile=public|private|domain[...]]
[type=static|dynamic]
[verbose]

Remarks:
- Displays all matching rules as specified by name and optionally,
  profiles and type. If verbose is specified all matching rules are
  displayed.

Examples:
Display all dynamic inbound rules:
netsh advfirewall firewall show rule name=all dir=in type=dynamic

Display all the settings for all inbound rules called
"allow browser":
netsh advfirewall firewall show rule name="allow browser" verbose

C:\Users\Asad>
  
```

Fig. — `netsh advfirewall firewall show rule name=all`

```

C:\Users\Asad>netsh advfirewall show currentprofile

Private Profile Settings:
-----
State: ON
Firewall Policy: BlockInbound,AllowOutbound
LocalFirewallRules: N/A (GPO-store only)
LocalGPOSecRules: N/A (GPO-store only)
InboundIdentifierNotification: Disable
OutboundIdentifierNotification: Disable
InboundResponseToMulticast: Disable
Logging:
LogAllInboundConnections: Disable
LogOutboundConnections: Disable
LogName: %Systemroot%\system32\Logfiles\Firewall\pfirewall.log
LogFileSize: 4096

Public Profile Settings:
-----
State: ON
Firewall Policy: BlockInbound,AllowOutbound
LocalFirewallRules: N/A (GPO-store only)
LocalGPOSecRules: N/A (GPO-store only)
InboundIdentifierNotification: Disable
OutboundIdentifierNotification: Disable
InboundResponseToMulticast: Disable
Logging:
LogAllInboundConnections: Disable
LogOutboundConnections: Disable
LogName: %Systemroot%\system32\Logfiles\Firewall\pfirewall.log
LogFileSize: 4096

OK.

C:\Users\Asad>
  
```

Fig. — `netsh advfirewall show currentprofile`

Result / Observation

The captured output above matches the expected output described for this command, confirming correct execution on the lab machine.

Chapter Summary

NETSH ADVFIREWALL is the command-line interface into the Windows Defender Firewall's profile and rule configuration.

5.14 NETSH WLAN

Purpose

Inspect wireless network adapter status, saved profiles, driver details, and supported wireless capabilities.

What the Command Does

NETSH WLAN reports on the current wireless interface state, previously saved Wi-Fi profiles, the installed wireless driver, and the security and band capabilities the adapter supports.

Syntax

```
netsh wlan show interfaces | show profiles | show drivers | show wirelesscapabilities
```

Important Options / Parameters

Option	Description
show interfaces	Current connection state, SSID, signal quality, and radio type.
show profiles	List of all saved wireless network profiles on the machine.
show drivers	Wireless adapter driver details and supported features.
show wirelesscapabilities	Security and band/standard capabilities of the adapter.

Example Command(s)

```
netsh wlan show interfaces
netsh wlan show profiles
netsh wlan show drivers
netsh wlan show wirelesscapabilities
```

Expected Output

show interfaces reports the connected SSID, signal percentage, radio type, and authentication/cipher in use; show profiles lists every saved network name; show drivers reports vendor, driver version, and supported radio types; show wirelesscapabilities lists supported authentication and encryption methods.

How to Read the Output

- Signal quality below roughly 50% often correlates with slow or unstable wireless performance.
- The Authentication/Cipher fields show whether the connection uses a secure protocol such as WPA2/WPA3-AES rather than an outdated one.
- A long list of saved profiles can represent a security exposure if old, unused networks are never removed.

Practical Use

- Confirm which wireless network the adapter is currently associated with and its signal strength.
- Review saved profiles when cleaning up a lab or shared machine.

Troubleshooting Use

- Diagnose weak-signal complaints using the reported signal percentage.
- Confirm the wireless driver is current and supports the required standard when troubleshooting slow Wi-Fi.

Security Relevance



SECURITY NOTE

Saved profiles for open or WEP-secured networks represent outdated, insecure configurations. Commands that reveal stored Wi-Fi keys should be restricted to authorized administrators only.

Common Mistakes



COMMON MISTAKE

Assuming a low signal percentage always means a hardware fault, rather than distance or interference. Leaving old, unused wireless profiles saved indefinitely on a shared machine.

Best Practice



BEST PRACTICE

Periodically review and remove unused saved wireless profiles. Confirm the adapter and driver support current security standards (WPA2/WPA3) before deploying it in production.

Practical Exercise

Run all four netsh wlan show commands and record the current SSID and signal quality, the number of saved profiles, the driver version, and the security methods the adapter supports.

Screenshot Evidence

[illegible]

Fig. — netsh wlan show interfaces

```
#Users\Aad Imannethalan show profiles

Profiles on interface Wi-Fi:

Group policy profiles (read only)

    <None>

User Profiles

All User Profile      : vivo V2le
All User Profile      : Siri Azam
All User Profile      : M-azam
All User Profile      : LBE-SAC-gen2-F4E2C68B8D8...
All User Profile      : Azam - Huteem
All User Profile      : Amir Alai
All User Profile      : Asad Isam
All User Profile      : Asad - Huteem
All User Profile      : MLMLK Ijaz
All User Profile      : eudorco
All User Profile      : Azam - Huteem
All User Profile      : M-azam
All User Profile      : Office
All User Profile      : Siri-Azam
```

Fig. — netsh wlan show profiles

[illegible]

Fig. — netsh wlan show drivers

[illegible]

Fig. — netsh wlan show wirelesscapabilities

Result / Observation

The captured output above matches the expected output described for this command, confirming correct execution on the lab machine.

Chapter Summary

NETSH WLAN provides full visibility into the wireless adapter's current connection, history, driver, and security capabilities.

6. Practical Exercises

The table below translates the original lab tasks into a consolidated, hands-on exercise list. Full guidance, syntax, and captured screenshot evidence for every one of these commands appears in its own chapter in Section 5.

#	Command(s)	What to Observe
1	ping <target>	Reachability, round-trip time, and packet loss to loopback, gateway, and a remote host.
2	ipconfig /all	IPv4/IPv6 address, subnet mask, gateway, DHCP status, and DNS servers.
3	getmac /v, hostname	The adapter's MAC address and the machine's configured hostname.
4	nslookup (+ -type=MX/NS/TXT)	Resolved address and record data from two different DNS servers.
5	tracert (+ -d)	Every hop and round-trip time between the host and destination.
6	pathping	Per-hop packet-loss percentage over an extended measurement window.
7	netstat -an/-ano/-r/-e/-s	Listening ports, established connections, PIDs, and protocol counters.
8	arp -a, arp -d *	Current IP-to-MAC mappings and how the cache repopulates after clearing.
9	systeminfo	OS version, boot time, memory, and adapter count.
10	route print	The default route, its interface, and its metric.
11	whoami, whoami /fqdn	Short vs. fully qualified session identity.
12	netsh advfirewall show ...	State of each firewall profile and total configured rules.
13	netsh wlan show ...	Current SSID/signal, saved profiles, driver, and security capabilities.
14	(analysis only)	Using the Exercise 5 trace, identify the first hop leaving the lab — the default gateway.

7. Screenshot-Based Documentation

Every command chapter in Section 5 includes a “Screenshot Evidence” subsection containing the actual Command Prompt output captured while performing that command on the lab machine, each labeled with the exact command executed. This provides a complete, verifiable visual record of every exercise alongside the analysis and explanation given in the surrounding text.

8. Professional Callout Boxes

Throughout this handbook, colored callout boxes are used consistently to highlight a specific type of information at a glance. The meaning of each callout used in this document is defined below.



TIP

A useful shortcut or practical piece of advice related to the surrounding topic.



BEST PRACTICE

A recommended, professional approach that should be followed in real-world practice.



COMMON MISTAKE

A frequent beginner error, included so it can be recognized and avoided.



IMPORTANT NOTE

An important concept that should be remembered and not overlooked.



SECURITY NOTE

A security implication relevant to the command or concept being discussed.



INTERVIEW TIP

Knowledge that is commonly useful in a technical interview setting.



THINK LIKE A NETWORK ADMINISTRATOR

A short prompt encouraging the reasoning habits of an experienced network administrator.



PRACTICAL SCENARIO

A realistic, educational situation illustrating when the command is useful in practice.

9. Real-World Scenarios

The scenarios below are educational and defensive in focus, showing how the commands in this handbook are used together by a network administrator or SOC analyst during real troubleshooting.

Scenario	Situation	Approach
PC cannot access the internet	Computer is powered on and connected but no websites load.	ipconfig /all → ping loopback → gateway → 8.8.8.8 → tracert to isolate where replies stop.
DNS resolution is failing	Sites work by IP but not by name.	nslookup against the default resolver and 8.8.8.8; flush the cache with ipconfig /flushdns and retest.
Device has an incorrect IP address	Host shows a 169.254.x.x (APIPA) address.	Confirm the physical link, then ipconfig /release followed by /renew to obtain a valid DHCP lease.
Network connection is slow	One application feels sluggish; browsing seems fine.	Baseline latency with ping, isolate the losing hop with pathping, confirm the session state with netstat.
Suspicious connection detected	Monitoring flags unusual outbound traffic from a host.	netstat -ano to find the owning PID, correlate in Task Manager, and check arp -a for local context.
Default gateway needs identifying	A new lab machine's path must be documented.	ipconfig /all and route print both show the gateway; the first tracert hop should match it.
Wireless adapter needs troubleshooting	A laptop repeatedly drops Wi-Fi in one area.	netsh wlan show interfaces for signal/radio type; show drivers to rule out an outdated driver.
Firewall configuration needs inspecting	An internal app can't be reached; a firewall block is suspected.	netsh advfirewall show allprofiles for state, then firewall show rule name=all for a missing allow rule.

10. Command Comparison Tables

The table below summarizes the primary purpose, network layer or area, and typical use of every command documented in this handbook.

Command	Main Purpose	Layer / Area	Typical Use
PING	Connectivity testing	Network (ICMP)	Reachability and latency check
IPCONFIG	IP configuration	Host	Configuration troubleshooting
GETMAC	Hardware addressing	Data Link	MAC address identification
HOSTNAME	Host identity	Host	Machine name verification
NSLOOKUP	DNS queries	DNS / Application	Name-resolution troubleshooting
TRACERT	Path discovery	Network	Route and hop analysis
NETSTAT	Connection analysis	Host / Transport	Connection and port troubleshooting
ARP	IP-to-MAC mapping	LAN / Data Link	Local network troubleshooting
PATHPING	Path + loss analysis	Network	Intermittent performance diagnosis
SYSTEMINFO	Host inventory	Host / OS	Asset and patch verification
ROUTE	Routing table management	Network	Path decision verification
WHOAMI	User identity	Host / Security	Session identity confirmation
NETSH ADVFIREWALL	Firewall management	Host / Security	Firewall state and rule audit
NETSH WLAN	Wireless management	Data Link / Physical	Wi-Fi status and security audit

11. Troubleshooting Workflow

The steps below present a complete, layered troubleshooting sequence built entirely from the commands documented in this handbook. Following this order avoids wasted effort by confirming the simplest, most local causes before investigating more distant or complex ones.

Step 1 — Check Physical / Network Connection

Before running any command, confirm the network cable is connected or the wireless adapter shows an active connection. No software diagnostic can compensate for a physical disconnection.

Step 2 — Check IP Configuration

```
ipconfig /all
```

Confirm the host has a valid IPv4 address, subnet mask, default gateway, and DNS servers. An APIPA address (169.254.x.x) or missing gateway points straight to a DHCP or configuration problem.

Step 3 — Test Localhost

```
ping 127.0.0.1
```

Confirms the local TCP/IP stack itself is initialized correctly, independent of any external network.

Step 4 — Test Default Gateway

```
ping <default gateway>
```

Confirms the host can reach the first device on its own local segment, isolating whether the problem is local or beyond the gateway.

Step 5 — Test Remote IP

```
ping 8.8.8.8
```

Confirms connectivity beyond the local network using a known-reachable public address, without involving DNS.

Step 6 — Test DNS Resolution

```
nslookup google.com
```

Confirms that name resolution itself is working, now that raw IP connectivity has already been proven.

Step 7 — Trace the Route

```
tracert <target>
```

If connectivity is only partially working or slow, traces the exact path and hop where the degradation begins.

Step 8 — Inspect Active Connections

```
netstat -ano
```

Confirms whether the expected application connections exist and identifies any unexpected or suspicious ones.

Step 9 — Inspect ARP Information

```
arp -a
```

Confirms local devices are resolving correctly at Layer 2 and helps rule out an IP or MAC address conflict.

Step 10 — Review Firewall / Wireless Configuration

```
netsh advfirewall show allprofiles
```

As a final step, confirms that a firewall rule or wireless configuration issue is not the underlying cause once basic connectivity has been ruled out.

12. Practical Analysis

Running a command is only half of the skill; the more valuable half is knowing what to look for in the result. A student should train themselves to scan output for the specific values listed below, rather than simply confirming that a command “ran successfully.”

What to Identify	Where It Appears	Why It Matters
IP address	ipconfig, ipconfig /all	Confirms the host's logical network identity.
Subnet mask	ipconfig /all	Defines the size of the local network segment.
Default gateway	ipconfig /all, route print	The exit point for all off-network traffic.
DNS server	ipconfig /all	Determines where name-resolution queries are sent.
MAC address	getmac /v, arp -a	The hardware identity used at Layer 2.
Open / listening ports	netstat -an	Reveals which services are exposed on the host.
Active connections	netstat -ano	Shows current communication with other hosts.
Routing information	route print, netstat -r	Shows how outbound traffic is directed.
ARP mappings	arp -a	Confirms local IP-to-MAC associations.
DNS records	nslookup	Confirms domain configuration (A, MX, NS, TXT).

What to Identify	Where It Appears	Why It Matters
Network hops	tracert, pathping	Shows the path and distance to a destination.
Packet loss	ping, pathping	Indicates congestion or an unreliable link.
Wireless adapter information	netsh wlan show interfaces	Shows current SSID, signal, and security in use.
Firewall profile / status	netsh advfirewall show allprofiles	Confirms whether the host is currently protected.



INTERVIEW TIP

Interviewers frequently ask candidates to “walk through” a troubleshooting scenario. Practicing the identification list above out loud, using this handbook's example commands, is excellent preparation.

13. Lab Results

Use the table below to record the outcome of each command exercise once it has been performed. Add additional rows as needed.

Command	Purpose	Observed Output	Key Finding	Result

(Add further rows as needed to cover all fourteen commands.)

14. Conclusion

This lab provided hands-on experience with the essential command-line utilities used to configure, inspect, and troubleshoot a Windows network connection. Beginning with basic connectivity testing through PING and progressing through IP configuration, DNS querying, path tracing, connection auditing, ARP inspection, system inventory, routing analysis, identity confirmation, and firewall/wireless review, this handbook has covered the complete toolkit a network administrator relies on for first-line diagnosis.

Beyond simply memorizing syntax, the goal of this lab was to build a layered, methodical way of thinking about network problems: confirm the local host first, then the immediate neighbor, then the wider path, and finally the name-resolution and security layers. This same sequence scales directly to real-world network administration, IT support, and security monitoring roles, where these exact commands are used daily to isolate outages, detect anomalies, and verify configuration changes.

Mastery of these fundamentals also forms the foundation for more advanced topics in later labs, including packet capture, protocol analysis, and network security monitoring, all of which build on the same core concepts of IP addressing, routing, and connection state introduced here.

15. Viva / Interview Questions

The following questions and short answers cover only the concepts and commands introduced in this lab and are suitable for viva voce preparation or general interview review.

Q1. What does the PING command test?

A: Basic IP-level reachability between a source and destination host using ICMP Echo Request and Echo Reply messages.

Q2. What does a high round-trip time in a ping reply indicate?

A: Increased latency, which may be caused by network congestion, distance, or an overloaded intermediate device.

Q3. Why might a host not respond to a ping even though it is online?

A: It may be configured to block or ignore ICMP Echo Requests, typically for security reasons.

Q4. What is the difference between ipconfig and ipconfig /all?

A: The plain form shows a brief summary (IP address, subnet mask, gateway); /all adds MAC address, DHCP status, lease times, and DNS servers.

Q5. What does an IP address in the 169.254.x.x range indicate?

A: An APIPA address, meaning the host could not reach a DHCP server and self-assigned an address.

Q6. What is the purpose of ipconfig /flushdns?

A: It clears the local DNS resolver cache so subsequent lookups query the DNS server fresh instead of using stale cached entries.

Q7. What information does GETMAC provide?

A: The physical (MAC) address of each network adapter installed on the computer.

Q8. Why can a MAC address not be used to route traffic across the internet?

A: MAC addresses are only meaningful on the local network segment; routers replace Layer 2 addressing at every hop.

Q9. What does the HOSTNAME command display?

A: The configured name of the local computer, used in local name resolution.

Q10. What does NSLOOKUP do?

A: It queries a DNS server directly to resolve a domain name to an IP address, or to retrieve a specific record type such as MX, NS, or TXT.

Q11. What does 'non-authoritative answer' mean in NSLOOKUP output?

A: The response came from a caching resolver rather than the domain's own authoritative name server.

Q12. What is an MX record used for?

A: It specifies the mail server(s) responsible for receiving email on behalf of a domain, along with a preference value.

Q13. How does TRACERT determine each hop along a path?

A: It sends packets with increasing TTL values, causing each router along the way to return a 'Time Exceeded' message that reveals its address.

Q14. Why might a hop in a tracert show only asterisks?

A: That specific router may be configured not to respond to ICMP, even though traffic is still passing through it successfully.

Q15. What additional information does PATHPING provide compared to TRACERT?

A: Quantified packet-loss percentages at each individual hop, gathered over an extended measurement period.

Q16. What does the NETSTAT -ano command show?

A: All active connections and listening ports, together with the numeric addresses and the owning process ID for each.

Q17. What does the state 'LISTENING' mean in NETSTAT output?

A: A local service is waiting for incoming connections on that port but has not yet established a session.

Q18. Why is NETSTAT useful in a security investigation?

A: It can reveal unexpected listening ports or outbound connections that may indicate malware or unauthorized software.

Q19. What does the ARP command display?

A: The local cache mapping recently contacted IP addresses to their corresponding MAC addresses.

Q20. What might it mean if two IP addresses map to the same MAC address in the ARP cache?

A: It may indicate ARP spoofing or a man-in-the-middle attack on the local network segment.

Q21. What kind of information does SYSTEMINFO provide?

A: OS version and build, system boot time, hardware details, memory, installed hotfixes, and basic network adapter information.

Q22. What does the 'route print' command show?

A: The local IP routing table, including destination networks, gateways, interfaces, and metrics.

Q23. What does the 0.0.0.0/0.0.0.0 entry in a routing table represent?

A: The default route, used for any destination that does not match a more specific route.

Q24. What is the difference between whoami and whoami /fqdn?

A: whoami shows the short DOMAIN\username identity; /fqdn shows the fully qualified domain name context of that identity.

Q25. Why do attackers commonly run 'whoami' after gaining access to a system?

A: To immediately confirm which user account and privilege level they have obtained.

Q26. What does netsh advfirewall show allprofiles display?

A: The state (ON/OFF) and default inbound/outbound policy of each Windows Firewall profile (Domain, Private, Public).

Q27. What does netsh wlan show interfaces report?

A: The currently connected SSID, signal quality, radio type, and the authentication/cipher in use.

Q28. What is the general order of steps in a layered network troubleshooting workflow?

A: Check the physical connection, verify IP configuration, test loopback, test the gateway, test a remote IP, test DNS, trace the route, and finally inspect connections, ARP, and firewall/wireless settings.

16. Quick Reference / Cheat Sheet

A consolidated quick-reference table of the major commands covered in this handbook.

Command	Purpose	Example
ping	Test connectivity to a host	ping 8.8.8.8
ipconfig /all	Show full IP configuration	ipconfig /all
ipconfig /release	Release the DHCP lease	ipconfig /release
ipconfig /renew	Request a new DHCP lease	ipconfig /renew
ipconfig /flushdns	Clear the DNS resolver cache	ipconfig /flushdns
ipconfig /displaydns	Show the DNS resolver cache	ipconfig /displaydns

Command	Purpose	Example
getmac /v	Show adapter MAC addresses	getmac /v
hostname	Show the local computer name	hostname
nslookup	Query DNS records	nslookup -type=MX gmail.com
tracert	Trace the path to a host	tracert 8.8.8.8
netstat -ano	List connections with PIDs	netstat -ano
netstat -r	Show the routing table	netstat -r
arp -a	Show the ARP cache	arp -a
arp -d *	Clear the ARP cache	arp -d *
pathping	Trace path with loss stats	pathping 8.8.8.8
systeminfo	Show system inventory	systeminfo
route print	Show the routing table	route print
whoami /fqdn	Show fully qualified identity	whoami /fqdn
netsh advfirewall show allprofiles	Show firewall profile states	netsh advfirewall show allprofiles
netsh wlan show interfaces	Show current Wi-Fi connection	netsh wlan show interfaces

=====Thank You=====